

Minor- III

School of Mathematics and Statistics
University of Hyderabad

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Duration: 60 minutes
Maximum Score: 20 points

Instructor: Dharmendra Kumar
Course: Eng Math - II

Instructions: You may use results proven in the lectures; however, answers without justification will receive a score of zero.

1. Let $f : \mathbb{C} \longrightarrow \mathbb{C}$.

- (a) Write the definition of singular point $z_0 \in \mathbb{C}$ of f .
- (b) Write the definition of an isolated singular point $z_0 \in \mathbb{C}$ of f .

[3 + 2]

2. Using the Cauchy's Residue Theorem, evaluate the integral

$$\int_C \frac{5z - 2}{z(z - 1)} dz$$

where C is the circle $|z| = 2$ described in counterclockwise.

[5]

3. If 20 identical notebooks are to be divided among 8 students. How many divisions are possible ? Justify your answer.

[5]

4. In a certain community, 36 percent of the families own a dog and 22 percent of the families that own a dog also own a cat. In addition, 30 percent of the families own a cat. What is the probability that a randomly selected family owns a dog given that it owns a cat ?

[5]

All the best !